

Final Module Assignment: Interactive Lecturing

Icebreaker: Two Truths and a Lie

Description of Activity

In my *Data Communications and Networking* class I use a version of the well known icebreaker Two Truths and a Lie as an interactive review tool. After completing a lecture (for example on IP addressing or the OSI model) I display three short statements related to the topic two accurate and one incorrect. Students must identify the lie. This playful approach transforms a traditional icebreaker into a content based formative assessment that reinforces key networking concepts.

Instructions for Use

- Present three statements related to the lecture topic.
- Students individually decide which statement is false.
- Facilitate a brief discussion on why they selected their answer.
- Reveal the correct answer and provide a concise explanation.

This can be done using PPTx, or Board during synchronous lectures.

Best Learning Environment

The activity works best in **synchronous environments** (in person or online sessions) because students can respond immediately and discuss reasoning. However in **asynchronous settings** it can be adapted into a discussion forum where learners post justifications for their answers.

Benefits to Learners

- Reinforces lecture material through immediate recall and reasoning.
- Encourages participation in a enjoyable environment.

Limitations or Barriers

- Time management may be challenging in longer lectures.
- Overuse can reduce novelty or engagement.

Incorporation into Class Activities

This activity is regularly used at the end of lectures in **CNET-184** as a knowledge check. It helps identify misconceptions early and keeps students attentive throughout class.

Active Learning Technique: Think Pair Share Debugging

Description of Activity:

Think-Pair-Share Debugging is an interactive learning technique grounded in discussion-based methods. It encourages students to identify and resolve errors collaboratively while explaining their reasoning. In CWEB-280 this involve analyzing short JavaScript or SQL code snippets.

Instructions for Use:

- Present students with a short piece of problematic code. Ask them to identify potential issues individually.
- Students discuss their reasoning in pairs or small groups, comparing problem-solving approaches.
- Each pair presents their reasoning to the class.

Learning Environments:

- Synchronous: Conduct live in class or breakout rooms using shared code editors
- Asynchronous: Students post their analysis of a code snippet on the course forum and respond

Benefits to Learners:

- Promotes deep understanding through verbalizing and peer teaching.
- Encourages critical thinking, communication, and collaborative problem-solving.

Limitations or Barriers:

- Some students may rely too heavily on peers rather than individual analysis.
- Requires carefully chosen examples to fit time limits.

Incorporation into Classroom Activity:

In CWEB this technique can be integrated during lessons on JavaScript functions or SQL joins